

Question Paper Preview

Notations :

- 1.Options shown in **green** color and with  icon are correct.
- 2.Options shown in **red** color and with  icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Industry 4.0

Section Id :	640653124560
Number of Questions :	23
Section Marks :	45
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531859623 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536025549.  YES

6406536025550.  NO

Question Id : 6406531859624 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (2 to 3)

Question Label : Comprehension

Sunrise Foods Pvt. Ltd. is a packaged snacks manufacturer located in Pune. The company uses a special type of corn starch as its key raw material for making chips. The production runs throughout the year, and maintaining a steady supply of raw materials is critical.

After reviewing the last year's consumption pattern, the Operations Manager found the following data:

- The company uses 24,000 kg of corn starch annually.
- The ordering cost (includes transportation, paperwork, inspection) is ₹250 per order.
- The annual holding cost (storage, insurance, spoilage risk) is ₹4 per kg per year.
- Corn starch is highly stable and does not deteriorate quickly.[Assume 52 weeks in an year]

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 2 Question Id : 6406531859625 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

What is the Economic Order Quantity (EOQ) in KG/order

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1700 to 1750

Question Number : 3 Question Id : 6406531859626 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

What is the total number of orders per year ?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

14

Question Id : 6406531859627 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (4 to 5)

Question Label : Comprehension

Bluescope company estimates its ordering cost at Rs 400 and carrying cost to be Rs.2/year/unit. The estimated annual requirement is 10,000 units at a unit price of Rs. 10.

Calculate the given subquestions:

Sub questions

Question Number : 4 Question Id : 6406531859628 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

Calculate the annual inventory carrying cost.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2000

Question Number : 5 Question Id : 6406531859629 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

While rechecked, Bluescope company finds that the order cost should have been 20% more than the currently estimated value. What should be the percentage increase in the EOQ in this case? Enter your answer without the '%' symbol.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

9.4 to 9.6

Question Id : 6406531859630 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (6 to 7)

Question Label : Comprehension

Suppose you need to accept reservation for a flight. There are total 100 seats in the aircraft. You have two classes of passengers - economy and business. One economy seat gets you 2,500 Rs of revenue while a business seat gets 7,500 Rs. From your experience, you know the demand for business travelers follows a uniform distribution in [10,20]. Then answer the given subquestions:

Sub questions

Question Number : 6 Question Id : 6406531859631 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

What is the Booking limit?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

83

Question Number : 7 Question Id : 6406531859632 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

What is the protection level?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

17

Question Number : 8 Question Id : 6406531859633 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Match the following columns:

Type of layout (Col A)	Description (Col B)
A1. Product Layout	B1. Equipment grouped by similar operations; high routing flexibility
A2. Process Layout	B2. Uses work cells for similar product families
A3. Fixed-Position Layout	B3. Equipment arranged according to product sequence
A4. Cellular Layout	B4. Work remains stationary; resources move to the job site

Options :

6406536025557. ✖ A1-B1, A2-B2, A3-B3, A4-B4

6406536025558. ✔ A1-B3, A2-B1, A3-B4, A4-B2

6406536025559. ✖ A1-B1, A2-B3, A3-B2, A4-B4

6406536025560. ✖ A1-B3, A2-B4, A3-B2, A4-B1

Question Number : 9 Question Id : 6406531859634 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Select the correct form of internet from column B corresponding to the example in Column A.

Column A	Column B
A1. Smart metering	B1. Consumer
A2. Banking system	B2. Commercial
A3. Entertainment integrated devices	B3. Industrial
	B4. Private

Options :

6406536025561. ✖ A1: B3, A2: B2, A3:B4

6406536025562. ✔ A1: B3, A2: B2, A3:B1

6406536025563. ✖ A1: B2, A2: B3, A3:B1

Question Number : 10 Question Id : 6406531859635 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

A factory works for two 8-hour shifts each day and produces a total of 24,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds. How many workstations are required to maintain the specified level of production (assuming that activities can be combined into those workstations in any order) (enter your answer in numeric value)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 11 Question Id : 6406531859636 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Calculate the Euclidean distance between two machines located at the coordinates A(5,10) and B(2,6) in a shop floor.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.8 to 5.3

Question Number : 12 Question Id : 6406531859637 Question Type : MSQ

Correct Marks : 1.5 Max. Selectable Options : 0

Question Label : Multiple Select Question

There are four demand points at A(1,2), B(2,3), C(3,5), D(4,1). The respective demand at these points are 7, 1, 3, 5. Select the optimal location based on cross-median approach

Options :

6406536025566. ✓ (2,2)

6406536025567. ✓ (3,2)

6406536025568. ✓ (2.5,2)

6406536025569. ✗ (2,3)

Question Number : 13 Question Id : 6406531859638 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following example(s) can be referred as seasonal pattern?

Options :

6406536025570. ✖ Economic recession

6406536025571. ✔ Increases Ceiling Fan sales during winters

6406536025572. ✔ Increased tourist footfall towards the end of the year

Question Number : 14 Question Id : 6406531859639 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following functions, Logistics Management does not cover?

Options :

6406536025573. ✖ Supply Management

6406536025574. ✖ Materials Management

6406536025575. ✔ Customer Management

6406536025576. ✖ Distribution Management

Question Number : 15 Question Id : 6406531859640 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the decision variable X_{ij} denotes in a Vehicle Routing Problem (VRP)?

Options :

6406536025577. ✔ If the vehicle travels node j from node i (binary)

6406536025578. ✖ Number of vehicles travelled between node i and node j (integer)

6406536025579. ✖ Weigh on the arc (i,j) e.g. the distance (integer)

Question Number : 16 Question Id : 6406531859641 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following model corresponds to Exponential Smoothing forecasting model?

Options :

6406536025580. ✔ $F_{t+1} = D_{t+1} - F_t$ where F_t, D_t represents the forecast and actual data for period t

6406536025581. ✖ $F_t = \mu - \theta * \epsilon_{t-1}$ where $\epsilon_{t-1} = D_{t-1} - F_{t-1}$

Question Number : 17 Question Id : 6406531859642 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following supply chain costs is/are affected by distribution network structure?

Options :

6406536025582. ✓ Inventory

6406536025583. ✓ Transportation

6406536025584. ✓ Facilities and handling

6406536025585. ✓ Information

6406536025586. ✗ Price

Question Number : 18 Question Id : 6406531859643 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is not an example of digitization?

Options :

6406536025587. ✗ Scanning a document to get the PDF document

6406536025588. ✗ Converting a physical 2D sheet to CAD drawing

6406536025589. ✗ Converting physical invoice to e-copy

6406536025590. ✓ Creating an automated report from the existing cost data

Question Id : 6406531859644 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (19 to 20)

Question Label : Comprehension

The Egg Dropping Problem is a Dynamic Programming (DP) problem that helps determine the minimum number of attempts needed to find the highest floor from which an egg can be dropped without breaking.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 19 Question Id : 6406531859645 Question Type : MCQ

Correct Marks : 1.5

Question Label : Multiple Choice Question

What is the state variable here?

Options :

6406536025591. ✓ Number of eggs on hand

6406536025592. ✗ Number of floors

6406536025593. ✗ Number of eggs that got broken

Question Number : 20 Question Id : 6406531859646 Question Type : MCQ

Correct Marks : 1.5

Question Label : Multiple Choice Question

How a stage is defines?

Options :

6406536025594. ✗ Number of eggs on hand

6406536025595. ✓ Number of floors

6406536025596. ✗ Number of eggs that got broken

Question Id : 6406531859647 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (21 to 22)

Question Label : Comprehension

Ramu puts up a stall for Diwali Light (Diya) every year. He builds a Diya at a cost of Rs. 3 and earns Rs.5 by selling each Diya. Any Diya not sold by Diwali will be discounted to Rs. 1. From his experience, Ramu knows that the demand is Uniformly Distributed within [800, 1000] units. Then answer the given sub-questions:

Sub questions

Question Number : 21 Question Id : 6406531859648 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the Critical Ratio?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Question Number : 22 Question Id : 6406531859649 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the optimal order quantity that minimizes Ramu's total cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

900

Question Number : 23 Question Id : 6406531859650 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the Industry 4.0 version of transportation.

Options :

6406536025599. ✖ IC engine

6406536025600. ✖ Electric car

6406536025601. ✔ Hyperloop

Question Number : 24 Question Id : 6406531859651 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Match Column A with Column B (country of origin)

Colum A	Column B
1. 1 st revolution	England
2. 2 nd revolution	USA
3. 3 rd revolution	USA/Japan
4. 4 th revolution	Germany

Options :

6406536025602. ✔ A1:B1, A2:B2, A3:B3, A4:B4

6406536025603. ✖ A1:B2, A2:B1, A3:B3, A4:B4

6406536025604. ✖ A1:B3, A2:B1, A3:B4, A4:B2

6406536025605. ✖ A1:B3, A2:B1, A3: B2, A4:B4

Question Number : 25 Question Id : 6406531859652 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements from below:

Options :

6406536025606. ✖ Optimal solution in a facility location problem using Euclidean method will result in a region

6406536025607. ✔ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity

6406536025608. ✔ Optimal solution in a facility location problem using Euclidean method will result in a point

6406536025609. ✖ Optimal solution in a facility location problem using Euclidean method matches the centre of gravity in most of the cases

Question Number : 26 Question Id : 6406531859653 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the demerits of outsourcing from the following list.

Options :

6406536025610. ✔ Reduced customer/supplier contact.

6406536025611. ✔ Loss of supply chain visibility.

6406536025612. ✔ Leakage of sensitive data and information.

6406536025613. ✔ Loss of internal capability and growth in third-party power.

Question Number : 27 Question Id : 6406531859654 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the name of the forecasting method represented by the following set of equations? The notations represent the usual meaning.

$$F_{t+1} = a_t + b_t$$

Where
$$a_t = \alpha D_t + (1 - \alpha)(a_{t-1} - b_{t-1});$$

$$b_t = \beta(a_t - a_{t-1}) + (1 - \beta)b_{t-1}$$

Options :

6406536025614. ✖ Exponential smoothing

6406536025615. ✖ Winter's model for seasonality

6406536025616. ✔ Holte's method for trend

6406536025617. ✖ None of these

Question Number : 28 Question Id : 6406531859655 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets in the neighbouring shelves. He picked a few milk-powder packets, however these are not part of his list. He was searching for Good-day biscuits. He could identify only one packet in the shelf remaining. However, there were lot of Sun-fist biscuits lying in the next shelf. So he picked ten Sun-fist packets. Now map the scenarios with the items below:

Product	Scenario
P1)Tea:Coffee	S1)Complementary
P2) Tea: milk-powder packet	S2)Substitution
P3) Good-day biscuits: Sun-fist bicuit	S3)Inventory dependant demand
	S4)Impulse buying

Options :

6406536025618. ✖ P1-S1, P2-S3, P3-S4

6406536025619. ✔ P1-S2, P2-S1, P3-S3

6406536025620. ✖ P1-S2, P2-S1, P3-S4